# Problem 1

Consider a three-link manipulator with the following DH table.

| Link | $a$ | $d$ | $\alpha$ | $\theta$ |
|---|---|---|---|---|
| 1 | 0 | $L_1$ | $\pi/2$ | $\star$ |
| 2 | $L_2$ | 0 | 0 | $\star$ |
| 3 | $L_3$ | 0 | 0 | $\star$ |

(a) Derive the forward kinematic equations symbolically for the position of the end effector in the base frame as a function of $\theta_1$, $\theta_2$, and $\theta_3$.
(b) Symbolically differentiate these kinematic equations to construct a symbolic representation of the position rows of the velocity Jacobian.
(c) Using the method outlined in class, calculate ${}^0\boldsymbol{z}_{k-1}$ and ${}^0\boldsymbol{d}_{k-1,N}$ for all $k = 1, \ldots, N$ symbolically and construct the position rows of the velocity Jacobian.
(d) For $L_1 = 1\,\text{m}$, $L_2 = 0.5\,\text{m}$, $L_3 = 0.3\,\text{m}$, $\theta_1 = 0\,\text{rad}$, $\theta_2 = \pi/4\,\text{rad}$, and $\theta_3 = \pi/4\,\text{rad}$, numerically evaluate your results from (b) and (c). *Hint: They should match!*

**Solution.** Abbreviations: $c_i = \cos\theta_i$, $s_i = \sin\theta_i$, $c_{23} = \cos(\theta_2 + \theta_3)$, $s_{23} = \sin(\theta_2 + \theta_3)$. With $\cos\frac{\pi}{2} = 0$ and $\sin\frac{\pi}{2} = 1$, the DH transform

$$\boldsymbol{A}_i = \text{Rot}_z(\theta_i)\,\text{Trans}_z(d_i)\,\text{Trans}_x(a_i)\,\text{Rot}_x(\alpha_i) = \begin{bmatrix} c_\theta & -s_\theta c_\alpha & s_\theta s_\alpha & a\,c_\theta \\ s_\theta & c_\theta c_\alpha & -c_\theta s_\alpha & a\,s_\theta \\ 0 & s_\alpha & c_\alpha & d \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

gives the link transforms

$${}^0\boldsymbol{T}_1 = \begin{bmatrix} c_1 & 0 & s_1 & 0 \\ s_1 & 0 & -c_1 & 0 \\ 0 & 1 & 0 & L_1 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad {}^1\boldsymbol{T}_2 = \begin{bmatrix} c_2 & -s_2 & 0 & L_2c_2 \\ s_2 & c_2 & 0 & L_2s_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad {}^2\boldsymbol{T}_3 = \begin{bmatrix} c_3 & -s_3 & 0 & L_3c_3 \\ s_3 & c_3 & 0 & L_3s_3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

**(a) Forward kinematics.** Multiplying the first two links,

$${}^0\boldsymbol{T}_2 = \begin{bmatrix} c_1c_2 & -c_1s_2 & s_1 & L_2c_1c_2 \\ s_1c_2 & -s_1s_2 & -c_1 & L_2s_1c_2 \\ s_2 & c_2 & 0 & L_1 + L_2s_2 \\ 0 & 0 & 0 & 1 \end{bmatrix},$$

and then ${}^0\boldsymbol{T}_3 = {}^0\boldsymbol{T}_1\,{}^1\boldsymbol{T}_2\,{}^2\boldsymbol{T}_3$ has the rotation block

$${}^0\boldsymbol{R}_3 = \begin{bmatrix} c_1c_{23} & -c_1s_{23} & s_1 \\ s_1c_{23} & -s_1s_{23} & -c_1 \\ s_{23} & c_{23} & 0 \end{bmatrix},$$

using $c_2c_3 - s_2s_3 = c_{23}$ and $c_2s_3 + s_2c_3 = s_{23}$. The end effector position is

$${}^0\boldsymbol{p} = {}^0\boldsymbol{d}_{0,3} = \begin{bmatrix} c_1\,(L_2c_2 + L_3c_{23}) \\ s_1\,(L_2c_2 + L_3c_{23}) \\ L_1 + L_2s_2 + L_3s_{23} \end{bmatrix}.$$

**(b) Differentiating the position.** The position rows of the Jacobian are $\boldsymbol{J}_v = \begin{bmatrix}\partial \boldsymbol{p}/\partial\theta_1 & \partial \boldsymbol{p}/\partial\theta_2 & \partial \boldsymbol{p}/\partial\theta_3\end{bmatrix}$:

$$\boldsymbol{J}_v = \begin{bmatrix} -s_1\,(L_2c_2 + L_3c_{23}) & -c_1\,(L_2s_2 + L_3s_{23}) & -L_3\,c_1s_{23} \\ c_1\,(L_2c_2 + L_3c_{23}) & -s_1\,(L_2s_2 + L_3s_{23}) & -L_3\,s_1s_{23} \\ 0 & L_2c_2 + L_3c_{23} & L_3\,c_{23} \end{bmatrix}.$$

**(c) Geometric method.** All joints are revolute, so column $k$ is ${}^0\boldsymbol{z}_{k-1} \times {}^0\boldsymbol{d}_{k-1,3}$. The joint axes are the third columns of ${}^0\boldsymbol{R}_{k-1}$:

$${}^0\boldsymbol{z}_0 = \begin{bmatrix}0\\0\\1\end{bmatrix}, \qquad {}^0\boldsymbol{z}_1 = {}^0\boldsymbol{z}_2 = \begin{bmatrix}s_1\\-c_1\\0\end{bmatrix}.$$

The vectors from each joint origin to the end effector are ${}^0\boldsymbol{d}_{k-1,3} = {}^0\boldsymbol{p}_{0,3} - {}^0\boldsymbol{p}_{0,k-1}$. Subtracting $L_1$ from the $z$-entry of ${}^0\boldsymbol{p}$ gives $\boldsymbol{d}_{1,3}$, and subtracting the origin of frame 2 gives $\boldsymbol{d}_{2,3}$:

$${}^0\boldsymbol{d}_{0,3} = \begin{bmatrix} c_1(L_2c_2 + L_3c_{23}) \\ s_1(L_2c_2 + L_3c_{23}) \\ L_1 + L_2s_2 + L_3s_{23} \end{bmatrix}, \quad {}^0\boldsymbol{d}_{1,3} = \begin{bmatrix} c_1(L_2c_2 + L_3c_{23}) \\ s_1(L_2c_2 + L_3c_{23}) \\ L_2s_2 + L_3s_{23} \end{bmatrix}, \quad {}^0\boldsymbol{d}_{2,3} = L_3 \begin{bmatrix} c_1c_{23} \\ s_1c_{23} \\ s_{23} \end{bmatrix}.$$

Taking the cross products column by column, with $\rho = L_2c_2 + L_3c_{23}$ and $h = L_2s_2 + L_3s_{23}$:

$$\boldsymbol{z}_0 \times \boldsymbol{d}_{0,3} = \begin{bmatrix} -s_1\rho \\ c_1\rho \\ 0 \end{bmatrix}, \quad \boldsymbol{z}_1 \times \boldsymbol{d}_{1,3} = \begin{bmatrix} -c_1h \\ -s_1h \\ \rho \end{bmatrix}, \quad \boldsymbol{z}_2 \times \boldsymbol{d}_{2,3} = L_3 \begin{bmatrix} -c_1s_{23} \\ -s_1s_{23} \\ c_{23} \end{bmatrix}.$$

Stacking these reproduces exactly the matrix found in (b).

**(d) Numerical check.** For $L_1 = 1\,\text{m}$, $L_2 = 0.5\,\text{m}$, $L_3 = 0.3\,\text{m}$, $\theta_1 = 0$, $\theta_2 = \theta_3 = \pi/4$: $c_{23} = 0$, $s_{23} = 1$, $\rho = 0.5\cos\frac{\pi}{4} = 0.3536$, $h = 0.5\sin\frac{\pi}{4} + 0.3 = 0.6536$, and ${}^0\boldsymbol{p} = [0.3536,\ 0,\ 1.6536]^\intercal\,\text{m}$. Both methods (Appendix A) give

$$\boldsymbol{J}_v^{(b)} = \boldsymbol{J}_v^{(c)} = \begin{bmatrix} 0 & -0.6536 & -0.3000 \\ 0.3536 & 0 & 0 \\ 0 & 0.3536 & 0 \end{bmatrix}\ \text{m/rad},$$

with a maximum entry-wise difference of 0, so the two results match.

# Appendix A: Code

## Problem1.m

Listing 1: Problem1.m

```
clear; clc;

%% Symbols and DH table (a, d, alpha fixed; theta is the joint variable)
syms th1 th2 th3 L1 L2 L3 real
th = [th1; th2; th3];
a       = [0;  L2; L3];
d       = [L1; 0;  0];
alphaDH = sym([pi/2; 0; 0]);   % exact symbolic so cos(pi/2) = 0

%% (a) Forward kinematics: 0T3 = A1*A2*A3
T = eye(4);
Ts = cell(1, 3);
for i = 1:3
    ct = cos(th(i)); st = sin(th(i));
    ca = cos(alphaDH(i)); sa = sin(alphaDH(i));
    A = [ct, -st*ca,  st*sa, a(i)*ct;
         st,  ct*ca, -ct*sa, a(i)*st;
         0,   sa,     ca,    d(i);
         0,   0,      0,     1];
    T = simplify(T * A);
    Ts{i} = T;                          % Ts{i} = 0T_i
end
p = simplify(T(1:3, 4));
disp('(a) Tool position 0p:'); disp(p);

%% (b) Differentiate the position equations
Jv_diff = simplify(jacobian(p, th));
disp('(b) Jv by differentiation:'); disp(Jv_diff);

%% (c) Geometric method: column k = z_{k-1} x d_{k-1,N}
z0 = sym([0; 0; 1]);
o  = {sym([0; 0; 0]), Ts{1}(1:3, 4), Ts{2}(1:3, 4)};    % o_0, o_1, o_2
z  = {z0, Ts{1}(1:3, 3), Ts{2}(1:3, 3)};                % z_0, z_1, z_2
Jv_geo = sym(zeros(3, 3));
for k = 1:3
    dk = p - o{k};                                       % 0d_{k-1,N}
    Jv_geo(:, k) = simplify(cross(z{k}, dk));
    fprintf('z_%d = ', k-1); disp(simplify(z{k}.'));
    fprintf('d_%d,3 = ', k-1); disp(simplify(dk.'));
end
disp('(c) Jv by z x d:'); disp(Jv_geo);
disp('Symbolic difference (b)-(c):'); disp(simplify(Jv_diff - Jv_geo));

%% (d) Numeric evaluation
vals = {L1, L2, L3, th1, th2, th3};
nums = {1, 0.5, 0.3, 0, pi/4, pi/4};
Jb = double(subs(Jv_diff, vals, nums));
Jc = double(subs(Jv_geo,  vals, nums));
```

```
pn = double(subs(p, vals, nums));
disp('(d) Op ='); disp(pn.');
disp('(d) Jv from (b) ='); disp(Jb);
disp('(d) Jv from (c) ='); disp(Jc);
fprintf('max |(b)-(c)| = %g\n', max(abs(Jb - Jc), [], 'all'));
```

## Appendix B: Hand Calculations

$C0 = 1$
$S0 = 0$

$C\frac{\pi}{2} = 0$
$S\frac{\pi}{2} = 1$

$\text{Rot}z(\theta) \cdot \text{Rot}x(\alpha) =$

$$\begin{bmatrix} C\theta & -S\theta C\alpha & S\alpha C\theta \\ S\theta & C\alpha C\theta & -S\alpha C\theta \\ 0 & S\alpha & C\alpha \end{bmatrix}$$

$$\begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \cdot p \qquad \begin{bmatrix} 0 & 0 & -C\theta_1 \\ 0 & 0 & -S\theta_1 \\ -C\theta_1 S\theta_1 & & 0 \end{bmatrix} {}^0\underline{d}_{1,3}$$

repeat

| L | a | d | $\alpha$ | $\theta$ |
|---|---|---|---|---|
| 1 | 0 | $L_1$ | $\pi/2$ | * |
| 2 | $L_2$ | 0 | 0 | * |
| 3 | $L_3$ | 0 | 0 | * |

$$J_v = \begin{bmatrix} {}^0\underline{V}_{0,3} \\ {}^0\underline{\omega}_{0,3} \end{bmatrix} \qquad \begin{matrix} 0 \\ 0 \\ 1 \end{matrix} \quad \begin{matrix} S\theta_1 \\ -C\theta_1 \\ 0 \end{matrix} \quad \begin{matrix} S\theta_1 \\ -C\theta_1 \\ 0 \end{matrix}$$

$$J_{v,lin} = \begin{bmatrix} {}^0\underline{V}_{0,3} \end{bmatrix} = \begin{bmatrix} {}^0\underline{z}_0 \times {}^0\underline{d}_{0,3} & {}^0\underline{z}_1 \times {}^0\underline{d}_{1,3} & {}^0\underline{z}_2 \times {}^0\underline{d}_{2,3} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{\partial P}{\partial \theta_1} & \frac{\partial P}{\partial \theta_2} & \frac{\partial P}{\partial \theta_3} \end{bmatrix}$$

$p$ $\quad {}^0p_{0,3} - {}^0p_{0,1} \quad {}^0p_{0,3} - {}^0p_{0,2}$

Sub $L_1$ from $z_0$

$$\begin{bmatrix} L_3 C\theta_1 C(\theta_2+\theta_3) \\ L_3 S\theta_1 C(\theta_2+\theta_3) \\ L_3 S(\theta_2+\theta_3) \end{bmatrix}$$

$$J_{v,lin} = \begin{bmatrix} -S\theta_1(C\theta_2 L_2 + C(\theta_2+\theta_3)L_3) & -C\theta_1(S\theta_2 L_2 + S(\theta_2+\theta_3)L_3) \\ C\theta_1(C\theta_2 L_2 + C(\theta_2+\theta_3)L_3 & -S\theta_1(S\theta_2 L_2 + S(\theta_2+\theta_3)L_3 \\ 0 & C\theta_2 L_2 + C(\theta_2+\theta_3)L_3 \end{bmatrix}$$

$$- \quad - \quad - \quad -$$

$$\begin{bmatrix} -C\theta_1 S(\theta_2+\theta_3)L_3 \\ -S\theta_1 S(\theta_2+\theta_3)L_3 \\ C(\theta_2+\theta_3)L_3 \end{bmatrix}$$

$$ {}^0T_1 = \begin{bmatrix} C\theta_1 & 0 & S\theta_1 & 0 \\ S\theta_1 & 0 & -C\theta_1 & 0 \\ 0 & 1 & 0 & L_1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$({}^0T_{1,3})$

$({}^1T_{2,3})$

$$ {}^1T_2 = \begin{bmatrix} C\theta_2 & -S\theta_2 & 0 & L_2 C\theta_2 \\ S\theta_2 & C\theta_2 & 0 & L_2 S\theta_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -S\theta_1 L_2 C\theta_2 - S\theta_1 L_3 C(\theta_2+\theta_3) & -C\theta_1 L_2 S\theta_2 + C\theta_1 L_3(-S\theta_2 C\theta_3 - C\theta_2 S\theta_3) & -L_3(C\theta_2 S\theta_3 - L_3 S\theta_2 C\theta_3) \\ C\theta_1 L_2 C\theta_2 + C\theta_1 L_3 C(\theta_2+\theta_3) & -S\theta_1 L_2 S\theta_2 + C\theta_1 L_3(-S\theta_2 C\theta_3 - C\theta_2 S\theta_3) & -L_3 C\theta_2 S\theta_3 - L_3 S\theta_2 C\theta_3 \\ 0 & L_2 C\theta_2 + L_3(C\theta_2 C\theta_3 - S\theta_2 S\theta_3) & -L_3 S\theta_2 S\theta_3 + L_3 C\theta_3 C\theta_3 \end{bmatrix} \begin{bmatrix} -C\theta_1 S(\theta_2+\theta_3)L_3 \\ -S\theta_1 S(\theta_2+\theta_3 L_3 \\ C(\theta_2+\theta_3)L_3 \end{bmatrix}$$

$$ {}^2T_3 = \begin{bmatrix} C\theta_3 & -S\theta_3 & 0 & L_3 C\theta_3 \\ S\theta_3 & C\theta_3 & 0 & L_3 S\theta_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$ {}^0T_2 = \begin{bmatrix} C\theta_1 C\theta_2 & -C\theta_1 S\theta_2 & S\theta_1 & L_2 C\theta_1 C\theta_2 \\ S\theta_1 C\theta_2 & -S\theta_1 S\theta_2 & -C\theta_1 & L_2 S\theta_1 S\theta_2 \\ S\theta_2 & C\theta_2 & 0 & L_1 + L_2 S\theta_2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$ {}^0T_3 = {}^0T_1 \, {}^1T_2 \, {}^2T_3$$

$$= \begin{bmatrix} C\theta_1(C\theta_2 C\theta_3 - S\theta_2 S\theta_3) & C\theta_1(-C\theta_2 S\theta_3 - S\theta_2 C\theta_3) & S\theta_1 & C\theta_1(L_2 C\theta_2 + L_3 C(\theta_2+\theta_3) \\ S\theta_1(C\theta_2 C\theta_3 - S\theta_2 S\theta_3) & S\theta_1(-C\theta_2 S\theta_3 - S\theta_2 C\theta_3) & -C\theta_1 & S\theta_1(L_2 C\theta_2 + L_3 C(\theta_2+\theta_3) \\ C\theta_2 S\theta_3 + S\theta_2 C\theta_3 & C\theta_2 C\theta_3 - S\theta_2 S\theta_3 & 0 & L_1 + L_2 S\theta_2 + L_3 S(\theta_2+\theta_3) \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$p$

$C\theta_2 C\theta_3 - S\theta_2 S\theta_3$

$S\theta_2 C\theta_3 + C\theta_2 S\theta_3$

